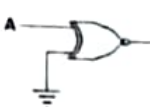
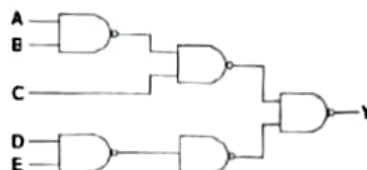


1.	In 2's complement representation the number 11100101 represents the decimal number																			
	A. +37	B. -31	C. +27	<u>D. -27</u>																
2.	For the gate in the given figure the output will be																			
																				
	A. 0	B. 1	C. A	<u>D. A'</u>																
3.	In the expression $A + BC$, the total number of minterms will be																			
	<u>A. 2</u>	B. 3	C. 4	<u>D. 5</u>																
4.	The circuit of the given figure realizes the function																			
																				
	<u>A. $Y = (A' + B')C + (DE)'$</u>		B. $Y = A' + B' + C' + D' + E'$																	
	C. $Y = AB + C + DE$		D. $Y = AB + C(D + E)$																	
5.	The number of product terms in the minimized sum of product expression obtained through the following K-map, where d, don't care.																			
	<table border="1" data-bbox="672 1142 911 1299"> <tr><td>1</td><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>d</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>0</td><td>d</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>0</td><td>1</td></tr> </table>				1	0	0	1	0	d	0	0	0	0	d	1	1	0	0	1
1	0	0	1																	
0	d	0	0																	
0	0	d	1																	
1	0	0	1																	
	<u>A. 2</u>	B. 3	<u>C. 4</u>	<u>D. 5</u> E. NONE																
6.	Minimum number of 2-input NAND gates required to implement the function $F = (x + y)(Z + W)$																			
	A. 3	B. 4	C. 8	<u>D. 6</u>																
7.	For the minterm designation $Y = \sum m(1, 3, 5, 7)$ the complete expression is																			
	A. $Y = A'B'C + AB'C$		B. $Y = A'B'C + AB'C + ABC + A'BC$																	
	C. $Y = A'B'C' + A'B'C + A'BC + AB'C$		<u>D. $Y = A'B'C' + ABC + A'B'C + AB'C$</u>																	
8.	The hexadecimal number $(3E8)_{16}$ is equal to decimal number																			
	<u>A. 1000</u>	B. 982	C. 768	D. 323																



9. For the K map in the given figure the simplified Boolean expression is /

CD \ AB	00	01	11	10
00				
01		1		
11		1	1	
10			1	

- A. $BCD + ACD$ ☒ B. $BC'D + ACD$
 C. $A'C + A'D' + ABC$ D. $A'C' + A'D' + ABC$

10. $AECF_{16} + 15ACD_{16} =$ /

- A. $A1111_{16}$ B. $A234F_{16}$ ☒ C. $C47BE_{16}$ D. $C47BB_{16}$

11. The minimum number of NAND gates required to implement the Boolean function $A + AB + ABC$ is equal to

- ☒ A. 7 B. 4 C. 1 D. 0

12. For the K map of the given figure, the simplified Boolean expression is /

CD \ AB	00	01	11	10
00	1	1		1
01	1	1		1
11			1	1
10				

- A. $AC'D + AC + BC$ B. $AC'D + AC$ C. $AB'D + BC$ ☒ D. $A'C' + A'D' + ABC$

13. In a 3 input NAND gate, the number of states in which output is 1 equals

- A. 6 B. 5 ☒ C. 1 D. 8

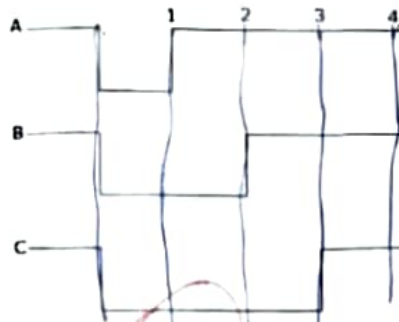


14. The inputs A, B, C of the given figure are applied to a 3 input NOR gate. The output is



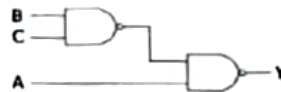
$$(A + B + C)'$$

$$A \cdot B' \cdot C'$$



- A. LOW from 0 to 2 and HIGH from 2 to 4
 B. HIGH from 0 to 1 and LOW from 1 to 4
 C. LOW from 0 to 4
 D. HIGH from 4 to 0

15. For the logic circuit of the given figure the simplified Boolean expression is



- A. $A + BC'$ B. $A' + B'C$ C. $A' + BC$ D. $A + BC$

16. In the circuit of the given figure, Y =

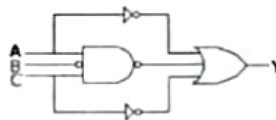


- A. X' B. X C. 1 D. 0

17. The complement of exclusive OR function is

- A. $AB + AB'$ B. $A'B + A$ C. $AB + A'B'$ D. $AB + A'B$

18. For the logic circuit of the given figure, the minimized expression is



- A. $Y = \overline{ABC}$ B. $Y = ABC$ C. $Y = A + B$ D. $Y = A + B + C$

19. 2's complement of (-24) =

- A. 00111111 B. 01111111 C. 01001000 D. 11101000

20. What is the result for expression (map method)?

$$(AB)'C + AD + D'(B + C) + AC' + (AD)'$$

- A. $A + D' + B'C$ B. 1 C. $A + D$ D. $A + D + B'C$

